

IOT BASED DRONE SYSTEMS

IV Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE08	ESC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
UNIT-I	INTRODUCTION TO IOT							
What is the IoT and why is it important? Elements of an IoT ecosystem, Technology drivers, Business drivers, Trends and implications, Overview of Governance, Privacy and Security Issues.								
UNIT-II	IOT ARCHITECTURE							
IOT ARCHITECTURE - IoT Open source architecture (OIC)- OIC Architecture &Design principles- IoT Devices and deployment models- IoTivity : An Open source IoT stack - Overview- IoTivity stack architecture- Resource model and Abstraction								
UNIT-III	WEB OF THINGS							
WEB OF THINGS - Web of Things versus Internet of Things – Two Pillars of the Web – Architecture Standardization for WoT– Platform Middleware for WoT – Unified Multitier WoT Architecture – WoT Portals and Business Intelligence								
UNIT-IV	UAV SUBSYSTEMS							
Mission Planning and Control Station- Types, Physical Configuration, Planning and Navigation, MPCs Interfaces, Modes of control, piloting and controlling mission, Autopilot system Launch Systems- Basic Considerations, launch Methods for Fixed, rotary wing UAV Recovery Systems								
UNIT-V	Interoperability in IoT and Introduction to Arduino Programming							
Sensing, Actuation, Basics of Networking, Communication Protocols ,Sensor Networks, Machine-to-Machine, Integration of Sensors and Actuators with Arduino								
Text Books:								
1. Honbo Zhou, “The Internet of Things in the Cloud: A Middleware Perspective”, CRC Press,2012. 2. Dieter Uckelmann, Mark Harrison, Michahelles, Florian (Eds), “Architecting the Internet of Things”, Springer, 2012.								
Reference Books:								
1. Cuno Pfister, Getting Started with the Internet of Things, O'Reilly Media, 2011, ISBN: 978-1-4493-9357-1 2. Skafidas, "Microcontroller Systems for a UAV", KTH, TRITA-FYS 2002:51 ISSN 0280-316X.34, 2002. 3.Vijay Madiseti and ArshdeepBahga, “Internet of Things (A Hands-on-Approach)”,1st Edition,VPT, 2014 4.Olivier Hersent, David Boswarthick, Omar Elloumi , “The Internet of Things – Key applications and Protocols”. Wiley, 2012								

COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:

1. Evaluate current trends in IoT and understand their implications on Aerospace industries and society.
2. Apply design principles to create IoT architectures considering various devices and deployment models.
3. Identify and utilize WoT portals effectively in various scenarios.
4. Analyze and propose suitable UAV subsystem configurations for specific missions.
5. Create basic IoT applications using knowledge acquired on concepts of interoperability in IoT, focusing on sensing, actuation, networking, and communication protocols